Task 1.1: Fully Visible Sigmoid Belief Network (FVSBN)

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1.1 Model · 1.2 Experimental setup · 1.3 Results · 1.4 Discussion · 1.5 Limitations

1.1 Model

The assignment asks for image generation with logistic regression, which is the fully visible sigmoid belief network. The idea is to write the joint distribution of an image as a product of one conditional per pixel,

$$p(x) \;=\; \prod_{i=1}^{784} p\big(x_i \mid x_{<i}\big), \tag{1.1}$$

where $x_{<i}$ are the pixels that come before i in a fixed generation order. For a binarised 28×28 image the vector has 784 entries and the natural order is row-major, $i = 28\,\mathrm{row} + \mathrm{col}$. Every conditional is a logistic regression on all previous pixels:

$$p\big(x_i = 1 \mid x_{<i}\big) \;=\; \sigma\Big(w_i^{\top} x_{<i} + b_i\Big), \qquad \sigma(z) = \frac{1}{1+e^{-z}}. \tag{1.2}$$

Stacking the 784 rows w_i into a matrix $W \in \mathbb{R}^{784\times 784}$ gives the vector form used in `fvsbn.py`:

$$\mathrm{logits}(x) \;=\; (W \odot M)^{\top} x + b, \qquad M_{ij} = \begin{cases} 1, & j < i, \\ 0, & \text{otherwise}, \end{cases} \tag{1.3}$$

where $\odot$ is the element-wise product, so row i of W only sees pixels with index below i. The model has $784 \times 784 + 784 = 615{,}440$ parameters and no hidden layer. It is called *fully visible* because the only latent quantity is the pixel being predicted: there is no latent variable that sums over the whole image, so nothing has to be approximated. Training is ordinary supervised learning on the observed pixels with the negative log likelihood

$$\mathcal{L}_{\mathrm{NLL}} \;=\; -\sum_{i=1}^{784} \Big[\, x_i \log p_i + (1 - x_i) \log(1 - p_i) \,\Big], \tag{1.4}$$

which is exactly 784 binary cross entropies added together, averaged over the batch. Because every conditional only depends on earlier pixels, the likelihood can be evaluated in one masked matrix product, and samples can be drawn ancestrally: draw x_1, then x_2 from $p(x_2 \mid x_1)$, and so on for 28 sequential steps.

1.2 Experimental setup

MNIST was binarised deterministically with a threshold of 0.5, so a pixel is 1 when it is brighter than 0.5. The fixed threshold keeps the training set identical between runs, which matters for reproducing the numbers. The resulting images contain 13.2% ink on average in the training split and 13.4% in the test split. The 60,000 training images are used for training; the 10,000 test images are used only for the reported numbers. The test NLL is also the validation signal during training, which is fine here because there is nothing to tune on it (the learning rate and batch size were fixed once, see below).

The optimiser is Adam with learning rate 10^{-3}, batch size 128 and 50 epochs, seed 0. There is no weight decay, dropout or learning rate schedule. An RTX 3070 Laptop GPU was used, and the same code runs on CPU. Two metrics are reported: the negative log likelihood per image (sum over the 784 pixels) and bits per dimension,

$$\mathrm{bpd} \;=\; \frac{\mathcal{L}_{\mathrm{NLL}}}{784 \ln 2}, \tag{1.5}$$

so lower is better for both and they are defined on the same binarised data.

Three ablations check that the numbers are not an accident of one setting: the number of training images (1k, 5k, 20k, 60k, 30 epochs), the binarisation threshold (0.3, 0.5, 0.7, 20k images, 30 epochs) and the pixel order (row-major versus column-major, 20k images, 30 epochs).

1.3 Results

The main run converges quickly and then flattens out (Figure 1.1). The first epoch already reaches 0.386 bpd, which is close to the unconditional baseline, and by epoch 10 the model is at 0.164. The final test numbers are 75.75 nats per image, 0.1394 bpd and 96.0% pixel accuracy under teacher forcing. Training took 30 to 50 s depending on GPU load (about 1 s per epoch, 60,000 images), and drawing 100 images takes about 0.06 s in the median of five runs, up to 0.15 s when the GPU is busy, so it is well under a millisecond per image despite the 28 sequential steps.

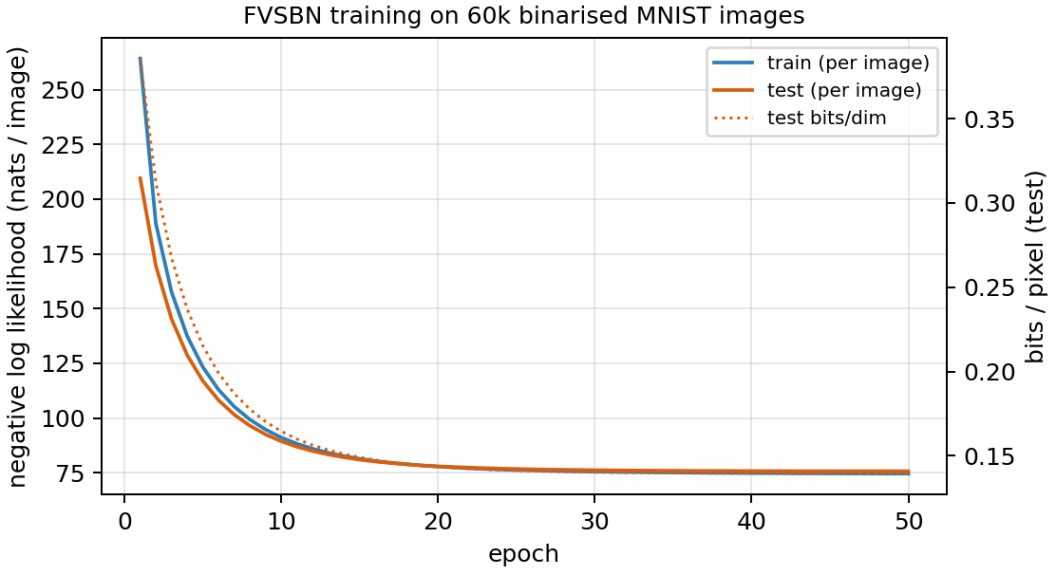


Figure 1.1: Training and test NLL per image; the right axis is the test bits per dimension.

The reference point is a model that ignores the prefix and predicts every pixel from its training marginal. That baseline costs 205.85 nats per image (0.3788 bpd), so the FVSBN removes about 63% of the baseline NLL. The gap between train and test NLL is small (74.6 versus 75.7 at the end), which says the linear model is underfitting rather than overfitting: more capacity, not more regularisation, is what this model needs.

Figure 1.2 shows four samples from the same model at epochs 1, 5, 10, 25 and 50. The first row is noise, by epoch 5 there are blobs, and from epoch 10 on the strokes of digits appear. Figure 1.3 shows 40 samples of the final model. The digits are recognisable but thick and sometimes smeared: the model reproduces the amount of ink (mean pixel 0.131 in the samples against 0.134 in the data) but not the fine structure, for example the thin diagonal of a 7 or the sharp corner of a 2.

Figure 1.4 looks at the same test digits in two ways. The middle column is the teacher-forced probability $p(x_i = 1 \mid x_{<i})$, which is mostly close to 0 or 1: given the true prefix the model is confident about the next pixel. The right column is the argmax of that probability, and it reconstructs the digit almost perfectly. So the per-pixel conditionals are good even though the joint samples are only rough. This is the usual exposure bias of autoregressive models: when sampling, the model conditions on its own earlier mistakes instead of the true pixels, and the small errors accumulate along the 784 steps.

The ablations are collected in Table 1.1 and Figure 1.5. The training set size is the dominant

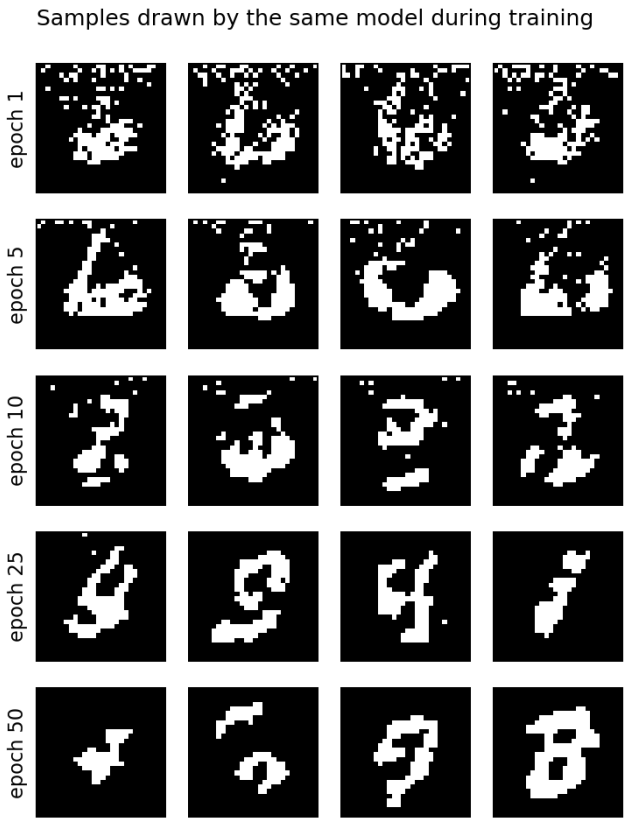


Figure 1.2: Samples drawn by the same model at epochs 1, 5, 10, 25 and 50.

factor: 1,000 images give 0.454 bpd, 5,000 give 0.289, 20,000 give 0.166 and 60,000 give 0.140. Most of the improvement happens before 20,000 images, and doubling to 60,000 only buys 0.015 bpd. The column-major order is slightly worse than row-major (0.1677 against 0.1656), which fits the fact that neighbouring pixels in a row are more correlated than pixels in a column for this kind of writing. The threshold rows need care: lowering the threshold to 0.3 adds ink and lowers bpd to 0.1604, raising it to 0.7 removes ink and gives 0.1717, but the data distribution is different in each case, so these three numbers only compare the model with itself at a fixed amount of ink.

Table 1.1: All runs, 30 epochs except the main run (50 epochs). Bits per dimension is $\mathcal{L}_{\mathrm{NLL}}/(784 \ln 2)$.

run	images	threshold	order	test NLL	test bpd
main	60000	0.5	row	75.75	0.1394
size 1k	1000	0.5	row	246.53	0.4537
size 5k	5000	0.5	row	156.94	0.2888
size 20k	20000	0.5	row	90.00	0.1656
size 60k	60000	0.5	row	76.19	0.1402
threshold 0.3	20000	0.3	row	87.19	0.1604
threshold 0.7	20000	0.7	row	93.32	0.1717
column-major	20000	0.5	col	91.14	0.1677

1.4 Discussion

The model does exactly what a logistic-regression generator can do. Every conditional is a linear function of the prefix, so the model can learn that a dark pixel is likely after a run of dark pixels, or



Figure 1.3: 40 samples of the final model.

that a curve continues in some direction, but it has no notion of a stroke being the same object at a different position. There is no weight sharing and no hidden unit that could detect a local pattern; each of the 784 rows of W has to learn on its own. That is also why it needs 615,440 parameters, about 4.5 times the number of values in a single image, and why increasing the training set still helps at 60,000 images.

The sampling experiments are the most interesting part of the section for the comparison with task 1.2. FVSBN reaches a decent bpd in about a minute of training, and the teacher-forced reconstructions are nearly perfect, but the joint samples are blurry. A hidden layer changes exactly this: it can represent non-linear interactions between pixels and reuse them at every position, which is the motivation for the neural autoregressive model in the next section. The course also covers estimators for models with latent units, but FVSBN does not need one: every variable is observed, so the likelihood is exact and the gradient is the gradient of a binary cross entropy.

Two smaller observations. First, the order of the pixels matters only a little in terms of NLL (0.1656 versus 0.1677), but it changes what a sample looks like: with column-major order the strokes are built from top to bottom, so errors in the first column are copied down the whole image. Second, the per-pixel conditionals are so confident that a naive argmax of the teacher-forced probabilities is a good reconstruction, which is why the model is sometimes described as a *deterministic* pixel predictor. The gap between that reconstruction and the samples is the price of sampling from the joint.

1.5 Limitations

The linear conditionals are the main limitation: for a given prefix the model predicts one Bernoulli probability per pixel, and a linear logit cannot describe curved or multi-modal conditional distributions. A second limitation is speed: sampling needs 28 sequential matrix products, so it is slow compared with a GAN generator that produces an image in one forward pass. A third is that the model is unconditional; generating a specific digit would need the label in the prefix. Adding a hidden layer (task 1.2) and conditioning on the class are the two obvious extensions, and the ablation table already shows where the gains would come from: more expressive conditionals, not more data.

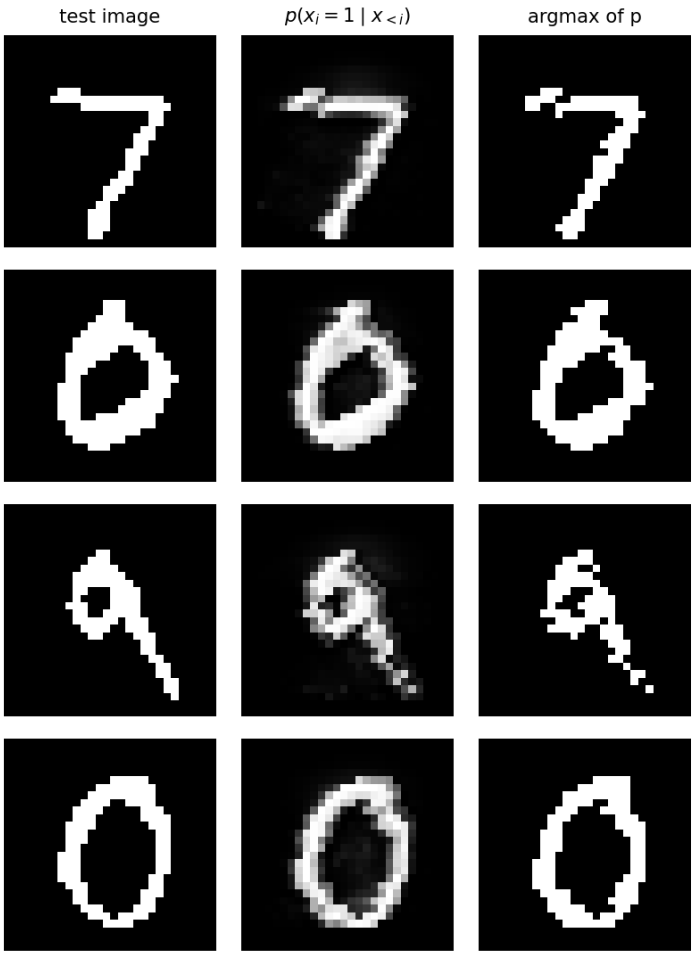


Figure 1.4: Test image, teacher-forced conditional probability and its argmax.

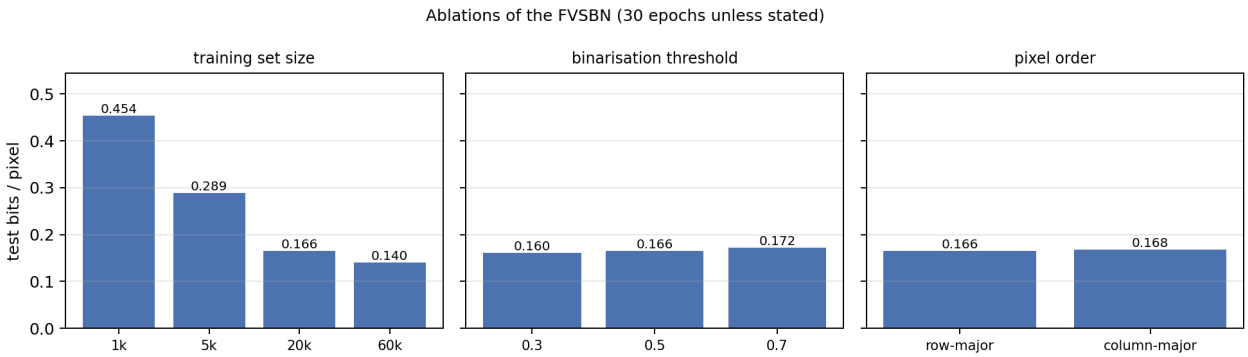


Figure 1.5: Ablations on training set size, binarisation threshold and pixel order.